K2: an arrival-aware compulsory merge

14 October 2026 — conditional candidate and interface gate, not adopted methods

1. Fixed geometry, information and the proposed handoff

Keep total road length L fixed. Candidate T1 reserves a terminal reach $b \in (0, L)$ and applies homogeneous early preparation on $[0, L_p]$, where $L_p = L - b$. An initial uniform footprint $[0, \ell]$, $0 < \ell \le L_p$, has ending, adjacent and other-continuing densities e_0, a_0, b_0, with $a_0 = b_0$ and $M = \ell(e_0 + a_0 + b_0) > 0$. The terminal reach is initially empty. There are no later arrivals or internal lateral sources in that reach. All quantities and the handoff are fixed across advice actions and comparators. The geometry and cessation of early transfer at L_p are proposals requiring adoption, not consequences of approving K2's review direction.

Use one common action $v \in [v_-, v_+]$, $0 < v_- < v_+$, and the H1 diagnostic family $k_c(v) = wk_j/(v + w)$, $C(v) = vk_c(v)$, with $w, k_j > 0$. Preserve the H1 initial bounds $0 \le e_0, a_0, b_0 \le k_c(v_+)$. Its homogeneous body evolves as

$$r = e_0 + a_0 = e(t) + a(t), \qquad \dot{e} = -\frac{v}{h} \min\left\{e, \frac{k_c(v) - r + e}{\gamma}\right\}, \quad a = r - e, \quad b(t) = b_0, \tag{1}$$

with fixed physical interaction scale $h > 0$ and early work weight $\gamma \ge 1$. The H1 proof of non-negative mass, common-speed free flow and lane-density invariance applies only while downstream admission does not hold the body. The adaptations in (1) remain as declared in the H1 note.

Let F_E, F_A, F_B be cumulative terminal-entry mass, initially zero. The preparation front and rear leave at $(L_p - \ell)/v$ and L_p/v. During that interval,

$$\dot{F}_E = ve(t), \qquad \dot{F}_A = va(t), \qquad \dot{F}_B = vb_0; \tag{2}$$

outside it all entry rates are zero. Early transfers into A are already in F_A and are not forced to merge a second time. Route-origin conservation may be checked by paired early transfers; service needs only the current completed/uncompleted class.

2. Physical stock, eligible queue and receiving at the interface

The terminal reach has free-flow delay $\tau = b/v$, backward delay $\tau_b = b/w$ and per-lane jam storage $H = k_j b$. Let Y_j be cumulative final departures for $j = E, A, B$. Extend initial histories as zero for negative time. The initially empty-link quantities are

$$N_j(t) = F_j(t) - Y_j(t), \quad P_j(t) = F_j(t - \tau), \quad q_j(t) = P_j(t) - Y_j(t). \tag{3}$$

N_j is physical stock; P_j is mass that could have reached the lane end without delay; q_j is eligible but unserved mass. These are different objects. A virtual eligibility time is not the physical moment a traveler joins the back of a spatial queue.

For $t > \tau_b$, the LTM vacancy is

$$V_j(t) = Y_j(t - \tau_b) + H - F_j(t). \tag{4}$$

Positive vacancy permits upstream receiving at capacity. At $0 < t \le \tau_b$, the empty-link vacancy is $k_j wt - F_j(t)$; capacity-bounded input makes it positive. The input rate must also respect $C(v)$. A conservative sufficient domain for all lanes and all actions is

$$M < H = k_j b, \qquad 0 \le \dot{F}_j(t) \le C(v). \tag{5}$$

Indeed $F_j \le M < H$ and $Y_j \ge 0$ ensure (4) is strictly positive after the backward delay. Before it, $F_j(t) \le C(v)t < k_j w t$. Thus terminal admission never restricts H1. This is a finite, deliberately conservative no-spillback domain, not an infinite buffer and not a test on q_E alone.

Eligible demand at the ending-lane exit is zero before τ. Afterwards it is its new eligible-arrival rate when $q_E = 0$ and its link capacity $C(v)$ when $q_E > 0$. This is the empty-link LTM queue/demand relation. The formulas are a restricted source-compatible reference; no assertion is made that the weighted node below equals Jin's general invariant junction model.

3. One queue, own-lane priority and finite forced service

Let $R > 0$ be declared constant available receiving-work rate for the adjacent exit and $R_B > 0$ the other exit's capacity. Ordinary mass uses one work unit; unfinished ending mass uses $\alpha \ge 1$. The late factor α is separate from γ. Select the elementary nominal domain

$$R > v_+ r, \qquad R_B \ge v_+ b_0. \tag{6}$$

This is stronger than merely fitting current continuing demand: the whole ordinary $E + A$ load fits R. Any forced queue arises from its extra receiving work, rather than an unacknowledged ordinary load excess. Source sending is respected because each lane's input is on the H1 free-flow branch.

Own-lane-first service then gives $Y_A = P_A$, $Y_B = P_B$ and $u_A = \dot{P}_A$. The remaining ending-lane service limit is

$$c_E(t; v) = \min\left\{C(v), \frac{R - u_A(t)}{\alpha}\right\} > 0. \tag{7}$$

The terminal flow and queue balance are

$$g_E(t) = \dot{Y}_E(t) = \begin{cases} c_E(t), & q_E(t) > 0, \\ \min\{\dot{P}_E(t), c_E(t)\}, & q_E(t) = 0, \end{cases} \qquad \dot{q}_E = \dot{P}_E - g_E. \tag{8}$$

Hence $q_E \ge 0$, $Y_E \le P_E$, g_E respects ending sending, and

$$u_A + \alpha g_E \le R, \qquad u_A + g_E \le R. \tag{9}$$

Transferred/departed mass is $g_E\,dt$, not $\alpha g_E\,dt$. Forced mass has no forward escape branch. Failed requests remain in (3), and the source link represents their slower congested motion implicitly. Advice is still the free-flow parameter; queued terminal traffic is not asserted to travel at that speed.

For the work-conserving diagnostic below, additionally require

$$R/\alpha \le C(v_-). \tag{10}$$

Since $C(v)$ increases with v, the longitudinal cap in (7) cannot bind in that subdomain. It is retained in the general candidate instead of hidden. Conditions (5), (6) and (10) are restrictions to qualify over the whole action range, not post-action receiving curves.

The entire finite cohort has become eligible by $t_* = L/v$. Thereafter own arrivals are zero and positive service is $\min\{C(v), R/\alpha\}$. Thus

$$T_{\text{clear}} = t_* + \frac{q_E(t_*)}{\min\{C(v), R/\alpha\}}, \tag{11}$$

with the no-queue case interpreted as $T_{\text{clear}} = t_*$. A positive native continuing density supplies the last physical tail; otherwise (11) is a finite upper bound. No unfinished mass is discarded at a horizon.

Source versus adaptation. Local own-lane-first and gap-weighted residual receiving are Carey ingredients. The compulsory boundary extension, constant terminal work factor/rate, restricted priority domain, final reach and lack of lateral sources there are explicit adaptations. Cooperative or proportional service is a different terminal rule, not a harmless replacement of (8); it can introduce an A queue. No field calibration or zipper law is claimed.

4. A joined domain with positive waiting and no upstream restriction

All the nominal restrictions can hold together. Choose positive sufficiently sparse initial densities so (5), (6) and (10) have room. At an admitted speed v_0, ending density at the first terminal arrival is positive because its finite H1 exposure has not annihilated all mass. Let those eligible densities be e_f, a_f, with $e_f + a_f = r$.

Positive initial queue growth occurs whenever

$$v_0 r < R < v_0(a_f + \alpha e_f) = v_0[r + (\alpha - 1)e_f]. \tag{12}$$

For $v_0 = v_+$ and $\alpha > 1$, this is a nonempty interval above the ordinary load bound. Sparse traffic can also make $R/\alpha < C(v_-)$ while $M < k_j b$. These are compatibility/existence inequalities, not fitted parameters or an interior-optimum certificate. Receiving at the upstream handoff remains slack under (5), even while the compulsory node has a queue under (12).

If (5) or the actual vacancy criterion fails, H1's prescribed entry clock is no longer trusted. If (6) fails, the one-queue restriction may no longer hold. Those failures reopen the smallest interface/domain contract; neither clipping incoming mass nor adding an undocumented second queue is permitted.

5. Complete TST, conservation and an arrival-pulse reference

All cohort mass is initially present at decision time. Total unexited mass and complete TST are

$$N_{\text{unexited}}(t) = M - Y_E(t) - Y_A(t) - Y_B(t), \qquad \text{TST} = \int_0^{T_{\text{clear}}} N_{\text{unexited}}(t)\, dt. \tag{13}$$

Paired early transfer conserves upstream mass, (3) conserves each terminal stream's input/output stock, and node discharge removes exactly the accepted mass. Gap work is never multiplied into population counts.

Because every initial position has free-flow eligibility time $(L - x)/v$ and only E is delayed in this domain,

$$\text{TST} = M\frac{L - \ell/2}{v} + \int_0^{T_{\text{clear}}} q_E(t)\, dt. \tag{14}$$

This is an identity for the admitted full trajectory. It counts physical travel through both regions and all excess terminal delay. It does not place a positive spatial queue after the road or claim that (3)'s eligible queue equals physical occupancy.

For an isolated node reference, let masses $m_A = m_E = 4$ arrive uniformly over duration d, set $R = 4$, $\alpha = 2$, and let sending be nonbinding. The reference clock starts at the first eligible arrival, not a newly chosen H1 decision boundary. During arrivals

$$u_A = m_A/d, \quad c_E = (R - m_A/d)/\alpha, \quad q_{\text{peak}} = \frac{m_A + \alpha m_E - Rd}{\alpha} \tag{15}$$

when the numerator is positive. If it is nonpositive, no queue forms. After arrivals the drain duration is $\alpha q_{\text{peak}}/R$. The two triangular queue areas sum to

$$\Delta\text{TST} = \frac{q_{\text{peak}} d}{2} + \frac{\alpha q_{\text{peak}}^2}{2R}. \tag{16}$$

For $d = 2$, $q_{\text{peak}} = 2$, drain duration is one, delay is three and the eligible-arrival-time sum $(m_A+m_E)d/2$ is eight: complete-time sum is eleven. For $d = 4$, the delay is zero and the complete-time sum is sixteen. These are closed-form hand references, not traffic rollouts, calibrated actions, or a reachable H1 comparison. Any whole-road prefix must still be included.

6. Priority reference: equal final work is not equal TST

Consider a different, already waiting batch with $m_A = m_E = 2$, $R = 2$ and $\alpha = 2$; all sending limits are slack. This comparison deliberately allows an A backlog and is outside the nominal one-queue input domain. Total work is six, so both following work-conserving allocations finish at time three.

Own-lane-first clears A at rate two over $[0, 1]$, then E at rate one over $[1, 3]$. The remaining population integrates to

$$\int_0^1 (4 - 2t)\, dt + \int_1^3 (3 - t)\, dt = 3 + 2 = 5. \tag{17}$$

Proportional work allocation maintaining the batch fractions serves both mass classes at rate $2/3$ over $[0, 3]$, and

$$\int_0^3 4(1 - t/3)\, dt = 6. \tag{18}$$

Thus S13's proportional batch is an actual conditional rule, not merely the name of a component. Its mean is not the mean of an own-lane-first queue. This reference neither calibrates priorities nor declares the cooperative alternative invalid. A full arrival-aware rival must specify its available demand, priority and own queue explicitly.

7. Fixed-preparation speed diagnostic: fastest can remain best

Hold completed preparation outcomes fixed across actions: each initial position carries the same terminal service class for every v. This is a deletion/reference case, not an assumption for the actual H1 sensitivity. For $v_f > v_s$, all free-flow eligibility times are earlier, so for all t

$$P_{A,f}(t) \geq P_{A,s}(t), \quad P_{E,f}(t) \geq P_{E,s}(t), \quad P_{B,f}(t) \geq P_{B,s}(t). \tag{19}$$

Assume the same fixed R, α and domain (5), (6), (10). Put

$$W(t) = P_A(t) + \alpha P_E(t), \qquad Z(t) = Y_A(t) + \alpha Y_E(t).$$

The weighted server is work-conserving: whenever weighted backlog is positive, it serves work at R. With zero initial eligible mass its cumulative output is

$$Z(t) = \inf_{0 \leq u \leq t} \{W(u) + R(t - u)\}. \tag{20}$$

This exact fluid reference follows from conservation and reflected queue service. Earlier cumulative input (19) implies $Z_f(t) \geq Z_s(t)$. Since $Y_A = P_A$, total completed mass is

$$Y_A + Y_E + Y_B = \frac{Z}{\alpha} + \left(1 - \frac{1}{\alpha}\right) P_A + P_B. \tag{21}$$

Every coefficient is nonnegative. Hence faster advice has at least as much completed mass at every time, and its integral (13) is no greater:

$$\mathrm{TST}(v_f) \le \mathrm{TST}(v_s). \tag{22}$$

Forced queueing alone therefore does not guarantee an interior minimum. Changed H1 preparation changes the input work/classes and invalidates the fixed-preparation premise of this proof. An actual interior optimum, if any, needs a full sensitivity/evidence pass and is not imposed as an acceptance criterion for this server.

As a separate deletion, setting $\alpha = 1$ under the ordinary-load bound (6) makes all arrivals fit the receiver and their own sending, so no ending queue forms. Complete time is then only the free-flow term in (14). This deletion does not claim aggregate equivalence in a broader domain that violates (6) or adds other receiver asymmetry, priorities or capacity-drop mechanisms.

8. Scientific disposition and references

Initial symmetry and an empty terminal mean M plus E_0 fixes $e_0 = E_0/\ell$, $a_0 = b_0 = (M - E_0)/(2\ell)$ and all candidate initial states. The strong aggregate-plus-ending-statistic rival can predict the same histories and queue. It is given all fixed parameters and the same observation contract. No future receiving oracle or extra initial lane observable is introduced.

The physical T1 construction and explicitly virtual T2 server are different methodological interpretations. T1 needs adoption of the terminal reach, early-transfer cutoff, proposed priority and joined domain; T2 needs acceptance of a deliberate no-feedback approximation. Neither exact law is adopted here. No applied joint calibration, reachable new information witness, AD/finite-difference evidence or independent usefulness result is claimed. The K2 pass stops at the original B2 interface gate; the modeling-chain stage remains open.

- Carey, Balijepalli and Watling (2013), Sections 5 / R1, R2 and Algorithms A1–A2: local receiving/priority ingredients. Institutional manuscript.
- Jin (2015), Sections 3.1–3.3, equations (23), (28)–(30): empty-link queue, vacancy and boundary counts. Primary preprint.
- Yperman, Logghe and Immers (2005), Sections 3–4: link/node decomposition and merge allocation. Original LTM paper.

All terminal geometry, no-lateral-source, weighted priority and action-domain adaptations are declared explicitly. The accepted S13 certificate remains unchanged.